HyperCLAP: Subspace Contrastive Learning Enables Compositional Audio Decomposition via Iterative Grassmannian Projection

A Proof of Concept

Likassi Régis[1]

[1]Independent Research

March 16, 2026

Abstract

Contrastive audio-language models such as CLAP and SLAP represent each audio signal as a single point in embedding space. We argue this is a geometric limitation: a point cannot simultaneously be close to two distant points, making mixture decomposition structurally impossible. We present **HyperCLAP**, a framework in which audio signals are embedded as low-dimensional subspaces on the Grassmann manifold $\mathrm{Gr}(k, d)$, with similarity measured by subspace overlap $\phi(S_i, S_j) = \|Z_i^\top Z_j\|_F^2$. Through a systematic experimental validation on ESC-50, we demonstrate three independent results: (1) a compositional training objective increases noise robustness monotonically by +49% as the compositional weight λ increases from 0 to 5 (Section 9); (2) an acoustic alignment phase teaches the encoder to map real acoustic mixtures to subspaces that overlap their constituent sources (pos_sim = 0.54 vs. neg_sim = 0.25, Section 11); (3) an iterative orthogonal projection algorithm on the Grassmannian recovers both source classes from a mixture in 65.2% of cases, compared to 12.0% for a point encoder baseline—a 5.4× improvement on a task that is *structurally impossible* with vector embeddings. These results constitute a proof of concept that subspace geometry provides a qualitatively new capability for audio representation learning.

1 Introduction

Contrastive learning has become the dominant paradigm for audio representation learning [Elizalde et al., 2023, Guinot et al., 2025]. These methods learn to align audio and text embeddings using InfoNCE-style objectives, collapsing rich acoustic scenes into single d-dimensional vectors.

This collapse is not merely a quantitative limitation—it is a *structural* one. A mixture of two audio sources $w_a + w_b$ produces a single point embedding $z_{\text{mix}} \in \mathbb{R}^d$. For retrieval to succeed, this point must be simultaneously close to z_a and z_b. By the triangle inequality, this is only possible if $z_a \approx z_b$—i.e., if the two sources are from the same class. For acoustically distinct sources, point-based mixture decomposition is geometrically prohibited.

Our hypothesis. Replacing point embeddings with k-dimensional subspaces $Z \in \mathbb{R}^{d\times k}$ on the Grassmann manifold $\mathrm{Gr}(k, d)$ resolves this structural limitation. A subspace can simultaneously overlap two distant subspaces, and—crucially—can be decomposed via orthogonal projection to extract each constituent source iteratively.

Contributions. This paper makes the following contributions:

1. We formalize the structural impossibility of mixture decomposition with point embeddings and show how subspace geometry overcomes it (Section 3).
2. We introduce a two-phase training procedure combining InfoNCE on pure sounds with an acoustic alignment loss on real waveform mixtures (Section 4).
3. We propose the **Iterative Grassmannian Decomposition** algorithm for zero-supervision source identification (Section 5).
4. We validate all three mechanisms independently on ESC-50 through a systematic experimental protocol (Sections 6 –7).

2 Background

2.1 Contrastive Audio-Language Models

CLAP [Elizalde et al., 2023] trains an audio encoder f_a and a text encoder f_t by maximizing the cosine similarity between matched audio-text pairs and minimizing it for unmatched pairs via the InfoNCE loss [van den Oord et al., 2018]: latex

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{e^{\cos(z_a, z_t^+)/\tau}}{e^{\cos(z_a, z_t^+)/\tau} + \sum_m e^{\cos(z_a, z_m^-)/\tau}}. \tag{1}$$

SLAP [Guinot et al., 2025] extends this to weakly supervised settings. Both methods produce point embeddings.

2.2 Grassmann Manifolds

The Grassmann manifold $\text{Gr}(k, d)$ is the space of all k-dimensional linear subspaces of $\mathbb{R}^d$. Each subspace S can be represented by an orthonormal basis $Z \in \mathbb{R}^{d\times k}$ with $Z^\top Z = I_k$. The canonical similarity between two subspaces is:

$$\phi(S_i, S_j) = \text{tr}(P_i P_j) = \|Z_i^\top Z_j\|_F^2 \tag{2}$$

where $P_i = Z_i Z_i^\top$ is the orthogonal projector onto S_i. This equals the sum of squared cosines of the principal angles between the subspaces, and lies in $[0, k]$.

2.3 Related Work

Subspace representations have been explored in face recognition [Basri and Jacobs, 2003], domain adaptation [Gopalan et al., 2011], and few-shot learning [Simon et al., 2020]. To our knowledge, HyperCLAP is the first work to apply Grassmannian geometry to audio-language contrastive learning.

3 Theoretical Analysis

Proposition 1 (Structural impossibility of point-based mixture decomposition)**.** *Let $z_a, z_b \in \mathbb{R}^d$ be point embeddings of two acoustically distinct sources with $\cos(z_a, z_b) < \delta$ for some small $\delta > 0$. Let z_{mix} be the embedding of their linear mixture $w_{\text{mix}} = \alpha w_a + (1-\alpha) w_b$. Then for any retrieval function r that returns the nearest neighbor in the corpus, $r(z_{\text{mix}})$ cannot simultaneously return both z_a and z_b unless $\cos(z_a, z_b) \geq 1 - \epsilon$ for negligible ϵ.*

Sketch. By the reverse triangle inequality for cosine similarity, any point equidistant from z_a and z_b lies on the bisector hyperplane, which is not guaranteed to be close to either source when they are geometrically distant. Top-1 retrieval returns a single point; correct dual-source identification requires the mixture representation to lie in the intersection of two distinct neighborhoods. □

Proposition 2 (Subspace compositionality)**.** *Let $S_a, S_b \in \text{Gr}(k, d)$ be subspaces with principal angles $\theta_1, \ldots, \theta_k$. The subspace $S_{\text{mix}} = \text{span}(Z_a \cup Z_b)$ satisfies $\phi(S_{\text{mix}}, S_a) > 0$ and $\phi(S_{\text{mix}}, S_b) > 0$ simultaneously, regardless of the geometry between S_a and S_b.*

Remark 1. *This is the key structural advantage. A $2k$-dimensional subspace can simultaneously overlap two k-dimensional subspaces even when those subspaces are orthogonal to each other. The Grassmannian provides the geometric degrees of freedom that $\mathbb{R}^d$ lacks for point embeddings.*

4 Method

4.1 Subspace Encoder

For each audio input x, the encoder $f : x \mapsto Z \in \mathbb{R}^{d\times k}$ produces an orthonormal basis via QR decomposition:

$$Z, _ = \text{QR}(g_\theta(x)), \quad g_\theta(x) \in \mathbb{R}^{d\times k}, \tag{3}$$

where g_θ is a CNN followed by a linear projection. This guarantees $Z^\top Z = I_k$ at every forward pass without explicit regularization.

4.2 Subspace InfoNCE Loss

We replace cosine similarity with the Grassmannian similarity $\phi(S_i, S_j)$ in the InfoNCE objective:

$$\mathcal{L}_{\text{sub}} = -\log \frac{\exp(\phi(S_i, S_i^+)/\tau)}{\exp(\phi(S_i, S_i^+)/\tau) + \sum_m \exp(\phi(S_i, S_m^-)/\tau)}. \tag{4}$$

4.3 Two-Phase Training

Phase 1 — Discriminative pre-training. We train the subspace encoder with $\mathcal{L}_{\text{sub}}$ on pure (unmixed) sounds for T_1 epochs. This produces semantically discriminative subspaces.

Phase 2 — Acoustic alignment. Given a real waveform mixture $w_{\text{mix}} = \alpha w_a + (1-\alpha) w_b$, we fine-tune the encoder to map w_{mix} to a subspace that overlaps both source subspaces. Source embeddings are frozen during this phase:

$$\mathcal{L}_{\text{align}} = \mathbb{E}\Big[\text{ReLU}\Big(\gamma + \phi_{\text{neg}}^{\text{hard}} - \tfrac{\phi(Z_m, Z_a) + \phi(Z_m, Z_b)}{2}\Big)\Big] \tag{5}$$

where $\gamma = 0.1$ is a margin and Z_n^{neg} are hard negatives sampled from classes other than a and b.

The frozen-source design is critical: it forces the encoder to learn an acoustic-to-subspace mapping for mixtures without distorting the already-learned representations of pure sounds.

5 Iterative Grassmannian Decomposition

At inference time, given a mixture embedding Z_{mix} and a corpus of reference embeddings $\{Z_1, \ldots, Z_N\}$, we recover the two constituent sources via orthogonal projection:

Algorithm 1 Iterative Grassmannian Decomposition

Require: $Z_{\text{mix}} \in \mathbb{R}^{d \times k}$, corpus $\mathcal{C} = \{Z_i\}_{i=1}^N$
Ensure: predicted source indices $(\hat{s}_1, \hat{s}_2)$
1: $\hat{s}_1 \leftarrow \arg\max_{i \in \mathcal{C}} \phi(Z_{\text{mix}}, Z_i)$
2: $P_\perp \leftarrow I_d - Z_{\hat{s}_1} Z_{\hat{s}_1}^\top$ *// orthogonal projector*
3: $Z_{\text{res}} \leftarrow \text{QR}(P_\perp Z_{\text{mix}})$ *// residual subspace*
4: $\hat{s}_2 \leftarrow \arg\max_{i \in \mathcal{C} \setminus \{\hat{s}_1\}} \phi(Z_{\text{res}}, Z_i)$
5: **return** $(\hat{s}_1, \hat{s}_2)$

Geometric interpretation. Step 2 computes the component of Z_{mix} orthogonal to the first source. The QR step in line 3 renormalizes the residual to remain on the Grassmannian. The second argmax then operates on the "source-1-free" subspace, which should be dominated by source 2.

No retraining required. Algorithm 1 is a purely geometric operation at inference time. It does not require any modification to the training procedure or any labeled mixture data.

6 Experiments

6.1 Dataset and Setup

We use **ESC-50** [Piczak, 2015], a benchmark of 2,000 environmental sound recordings across 50 balanced classes (44.1 kHz, 5-second clips). For mixture experiments, we use a reduced corpus of 10 classes (400 clips) to make the retrieval task tractable.

Architecture. Both subspace and point encoders share the same CNN backbone: three convolutional blocks with BatchNorm and GELU activations, followed by adaptive average pooling and a two-layer linear projection head. The subspace encoder projects to $\mathbb{R}^{d \times k}$ and applies QR; the point encoder projects to $\mathbb{R}^d$ and applies ℓ_2 normalization. We use $d = 64$, $k = 4$, $\tau = 0.1$, batch size 32, Adam optimizer with $\text{lr} = 3 \times 10^{-4}$.

Audio preprocessing. Log-mel spectrograms: $n_{\text{mels}} = 128$, $n_{\text{fft}} = 1024$, hop $= 512$. Temporal dimension pooled to $T = 128$.

6.2 Experimental Protocol

We follow a systematic validation protocol across 11 sections:

1. **S5**: PCA prototype (structural baseline)
2. **S6**: Learned encoders, pure InfoNCE
3. **S7**: Augmentation robustness (standard)
4. **S8**: Mixture-aware contrastive training
5. **S9**: λ ablation (causal test)
6. **S10**: Direct mixture retrieval (instance-level)
7. **S11**: Acoustic alignment + iterative decomposition

6.3 Metrics

Noise robustness (S7–S9): mean cosine/subspace similarity between a clip and its noise-augmented version (SNR $\in [10, 30]$ dB).

Recall@k (S10–S11, class-level): fraction of mixture queries for which both source classes appear in the top-k retrieved classes.

Both-class accuracy: fraction of queries where both source classes are correctly identified (class-level Recall@2).

7 Results

7.1 S8–S9: Compositional Training and Noise Robustness

Table 1 reports noise robustness as a function of the compositional loss weight λ. The monotonic increase confirms that the compositional loss is the causal mechanism, not the architecture.

The +49.3% total gain from $\lambda = 0$ to $\lambda = 5$ without any sign of saturation suggests that the Grassmannian geometry acts as an implicit regularizer: the compositional constraint forces the k basis vectors to capture invariant directions that are robust to additive perturbations.

Table 1: Noise robustness vs. compositional loss weight λ. $\lambda = 0$ is equivalent to standard InfoNCE. All models use the same architecture.

λ	Noise similarity ↑	Δ vs. $\lambda = 0$
0.0	0.6490	—
0.5	0.7920	+0.143
1.0	0.8070	+0.158
2.0	0.8951	+0.246
5.0	0.9688	+0.320

7.2 S11: Iterative Grassmannian Decomposition

Table 2 compares three methods on the class-level mixture decomposition task (10-class corpus, 400 clips).

The iterative decomposition achieves 65.2% both-class accuracy, compared to 12.0% for the point encoder—a **5.4**× improvement. Notably, both baselines perform at or below the random baseline, confirming the structural impossibility described in Section 3: mixture embeddings fall in a region of the space orthogonal to their sources in the absence of geometric decomposition.

The constant Recall@k across all k values for the decomposition method reflects the binary nature of the algorithm (it either returns both classes or neither), which is a consequence of the hard argmax in Algorithm 1.

7.3 Phase 2 Training Dynamics

After acoustic alignment (Phase 2, 40 epochs), the encoder achieves pos_sim = 0.54 (similarity between mixture embedding and its source embeddings) versus neg_sim = 0.25 (similarity to negative sources). This 2.2× ratio confirms that the encoder has learned to place mixture embeddings geometrically closer to their constituent sources than to unrelated sounds.

8 Discussion

8.1 Why Subspaces Enable What Points Cannot

The key insight is geometric rather than algorithmic. A k-dimensional subspace has k degrees of freedom to allocate across multiple directions. When trained with a compositional objective, the encoder distributes these directions to capture multiple concurrent sources. An orthogonal projection then surgically removes one source's contribution, exposing the other in the residual subspace.

This operation has no analogue in point-embedding spaces. The residual of a vector after subtracting a direction is another vector, but it carries no guarantee of semantic coherence—the subtracted direction is not interpretable as a source representation.

8.2 Implications for Audio-Language Learning

Our results suggest that the standard CLAP paradigm has a structural ceiling for compositional tasks. Replacing point embeddings with subspace embeddings is not an incremental improvement—it enables a qualitatively new inference mode. We identify several immediate applications:

Unsupervised source separation. Given a mixed recording and a corpus of reference sounds, Algorithm 1 identifies constituents without any separation supervision.

Compositional audio search. Queries of the form "find audio containing both rain and birds" can be answered by checking subspace overlap with each query component independently.

Biodiversity monitoring. Ecological recordings contain simultaneous species vocalizations. Iterative decomposition can identify each species sequentially from a single recording.

Industrial anomaly detection. Normal machine noise and anomalous events co-occur; decomposition isolates the anomaly.

8.3 Limitations

Scale. All experiments use a 400-clip corpus. Validation on AudioSet [Gemmeke et al., 2017] (2M clips, 527 classes) is necessary to establish practical viability.

Architecture. Results use a small CNN backbone. Whether the geometric advantage persists with large pre-trained encoders (HTSAT, BEATs) is an open question.

Decomposition accuracy. The 65.2% both-class accuracy leaves 34.8% failures. These likely correspond to acoustically similar source pairs where the residual subspace retains significant energy from the first source.

Runtime. QR decomposition at each forward pass adds computational overhead proportional to $O(dk^2)$ per sample.

9 Conclusion

We presented HyperCLAP, a proof of concept demonstrating that Grassmannian subspace representations

Table 2: Class-level mixture decomposition on ESC-50 (10 classes, 400 clips, 399 queries). Both-class accuracy: fraction of queries where both source classes are correctly identified.

Method	@1	@2	@5	@10	Both@2
Point encoder (baseline)	0.085	0.120	0.148	0.173	0.120
HyperCLAP direct	0.085	0.113	0.160	0.213	0.113
Ours (decomp.)	**0.652**	**0.652**	**0.652**	**0.652**	**0.652**
Random baseline	0.100	0.200	0.500	1.000	0.200

enable compositional audio decomposition that is structurally impossible with point embeddings. Three independent experimental results support this claim:

1. Compositional training increases noise robustness by +49% monotonically with λ, establishing a causal link between the training objective and robustness.
2. Acoustic alignment training produces mixture embeddings with 2.2× higher similarity to their sources than to negative sounds.
3. Iterative orthogonal projection on the Grassmannian identifies both source classes in 65.2% of mixture queries, against 12.0% for a point encoder baseline—a 5.4× improvement on a task that is structurally impossible for point-based models.

These results motivate a full-scale validation of HyperCLAP on large audio-language benchmarks and open a new geometric direction for multimodal representation learning.

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